OXFORD AQA
INTERNATIONAL QUALIFICATIONS

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname ______________________

Forename(s) ______________________

Candidate signature ______________________
I declare this is my own work.

INTERNATIONAL GCSE
CHEMISTRY
Paper 2

Thursday 9 November 2023 07:00 GMT Time allowed: 1 hour 30 minutes

Materials
For this paper you must have:
- a pencil and a ruler
- a scientific calculator
- the periodic table (enclosed).

Instructions
- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- Show all your working.

Information
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 90.
- You are expected to use a scientific calculator where appropriate.
- A periodic table is provided as a loose insert.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	

Do not write outside the box

Answer **all** questions in the spaces provided.

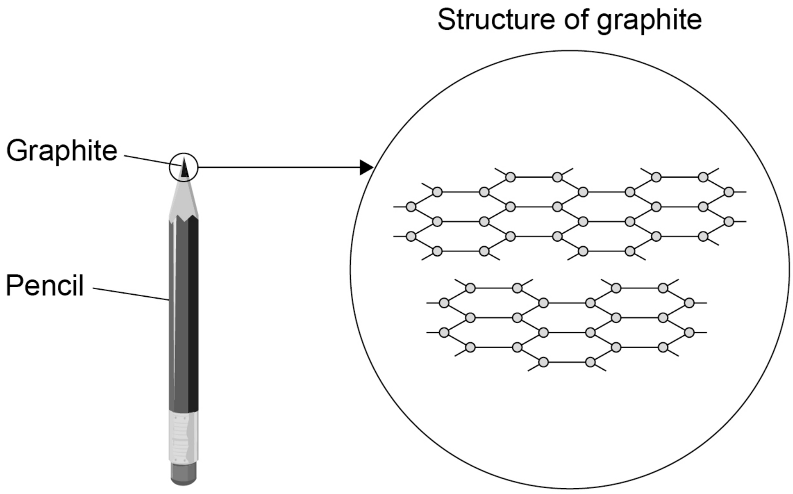
0 1 This question is about forms of carbon.

Graphite is a form of carbon.
Graphite is used in pencils.

Figure 1 shows:

- a pencil containing graphite
- a representation of the structure of graphite.

Figure 1



0 1 . 1 Which type of bonding is present in graphite?

[1 mark]

Tick (✓) **one** box.

Covalent ☐

Ionic ☐

Metallic ☐

Do not write outside the box

0 1 . 2 How many bonds does each carbon atom in graphite form?

Use **Figure 1**.

[1 mark]

Tick (✓) **one** box.

1 ☐

2 ☐

3 ☐

4 ☐

0 1 . 3 Why is graphite used in pencils?

[1 mark]

Tick (✓) **one** box.

Graphite has a high melting point ☐

Graphite is insoluble ☐

Graphite is slippery ☐

0 1 . 4 Explain why graphite conducts electricity.

You should refer to electrons in your answer.

[2 marks]

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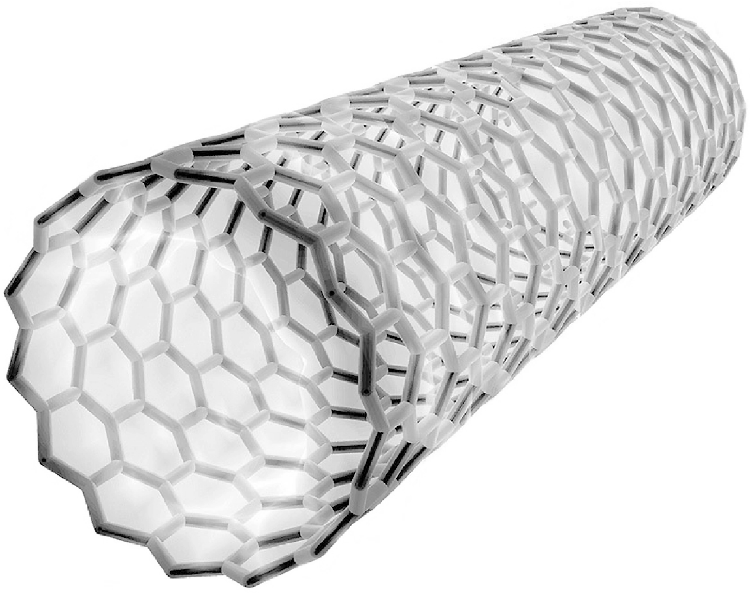
Question 1 continues on the next page

Fullerenes are also forms of carbon.

Figure 2 represents the structure of a fullerene.

Figure 2



0 1 . 5 Give **two** uses of fullerenes.

[2 marks]

1 ____

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2 ____

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Do not write outside the box

0 1 . 6 Fullerenes are nanoparticles.

What is the size of a nanoparticle?

[1 mark]

Tick (✓) **one** box.

1–100 nm ☐

1–100 mm ☐

1–100 cm ☐

1–100 m ☐

8

Turn over for the next question

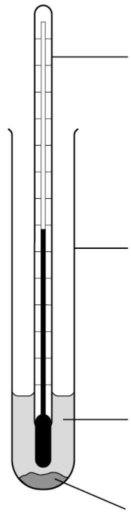
Do not write outside the box

0 2 A student investigated the reactivity of different metals.

The student determined the temperature rise when each metal was added to copper sulfate solution.

Figure 3 shows the apparatus.

Figure 3



This is the method used.

1 Add 10 cm^3 of copper sulfate solution to the boiling tube.

2 Measure the temperature of the copper sulfate solution.

3 Add 0.5 g of magnesium powder to the copper sulfate solution.

4 Measure the highest temperature reached.

5 Repeat steps **1** to **4** twice more.

6 Repeat steps **1** to **5** using different metal powders.

0 2 . 1 What is the independent variable in this investigation?

[1 mark]

Tick (✓) **one** box.

Temperature rise ☐

Type of metal ☐

Volume of copper sulfate solution ☐

0 2 . 2 **Table 1** shows the results.

Table 1

	Temperature rise in °C			
	1	**2**	**3**	**Mean**
Iron	4.2	4.4	4.0	4.2
Magnesium	10.4	10.2	10.5	10.4
Zinc	5.3	5.7	5.6	**X**

Calculate the mean temperature rise (**X**) for zinc metal.

Give your answer to 2 significant figures.

[3 marks]

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Mean temperature rise **X** (2 significant figures) = ______________ °C

0 2 . 3 Which is the most reactive metal?

Give **one** reason for your answer.

Use **Table 1**.

[2 marks]

Metal ______________________________

Reason ______________________________

____

Question 2 continues on the next page

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0 2 . 4 The student changed the investigation to improve the accuracy of the results.

Which change would improve the accuracy of the results?

[1 mark]

Tick (✓) **one** box.

Add insulation to the boiling tube ☐

Increase the volume of the copper sulfate solution ☐

Use a thermometer with a resolution of 1 °C ☐

0 2 . 5 Name the **two** products formed when magnesium reacts with copper sulfate solution.

[2 marks]

1 ____________________

2 ____________________

9

Do not write outside the box

0 3 This question is about methane (CH_4).

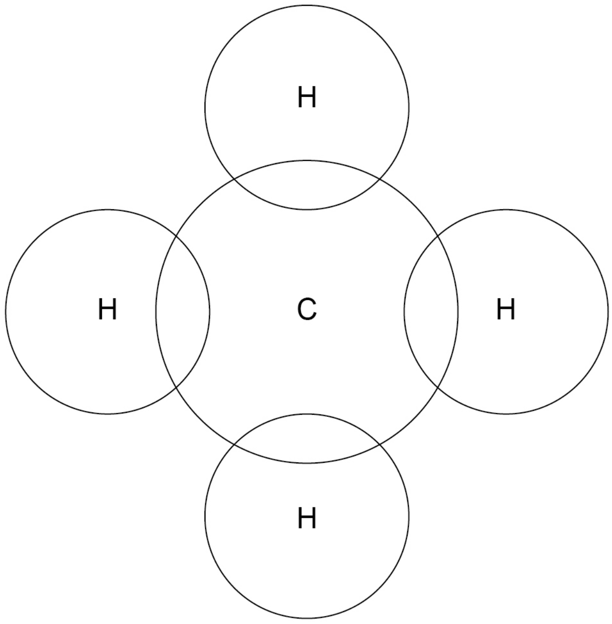
0 3 . 1 Methane is formed from carbon and hydrogen atoms.

A carbon atom has 4 outer electrons and a hydrogen atom has 1 outer electron.

Complete **Figure 4** to show the arrangement of **outer shell** electrons in methane.

Use dots (o) and crosses (x) to represent the electrons.

Figure 4



[2 marks]

Question 3 continues on the next page

Methane is a simple molecule.

Table 2 shows the electrical conductivity of three substances **A**, **B** and **C**.

Table 2

Substance	**Electrical conductivity when solid**	**Electrical conductivity when liquid**
A	Good	Good
B	Poor	Poor
C	Poor	Good

0 3 . 2 Which substance consists of simple molecules?

Give **one** reason for your answer.

Use **Table 2**.

[2 marks]

Substance ________________

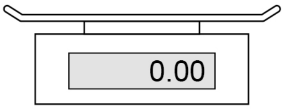
Reason ____

____

0 3 . 3 Which apparatus can be used to determine the results shown in **Table 2**?

[1 mark]

Tick (✓) **one** box.



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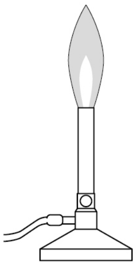

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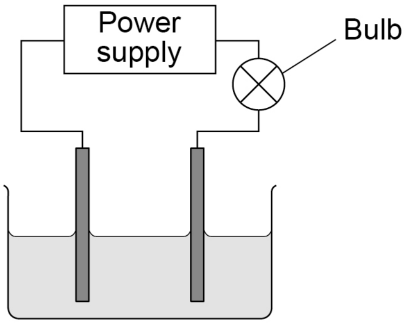

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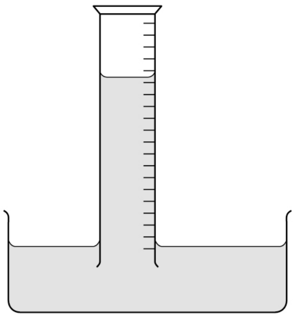

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Question 3 continues on the next page

Methane, ethane and propane are all simple molecules.

Table 3 shows information about methane, ethane and propane.

Table 3

Name of molecule	Number of carbon atoms in a molecule	Boiling point in °C
Methane	1	–161.5
Ethane	2	–88.5
Propane	3	–42.0

0 3 . 4 Complete the sentence.

[1 mark]

As the number of carbon atoms in a molecule increases, the boiling point

________________________________.

0 3 . 5 Explain why methane, ethane and propane have low boiling points.

[2 marks]

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8

Turn over for the next question

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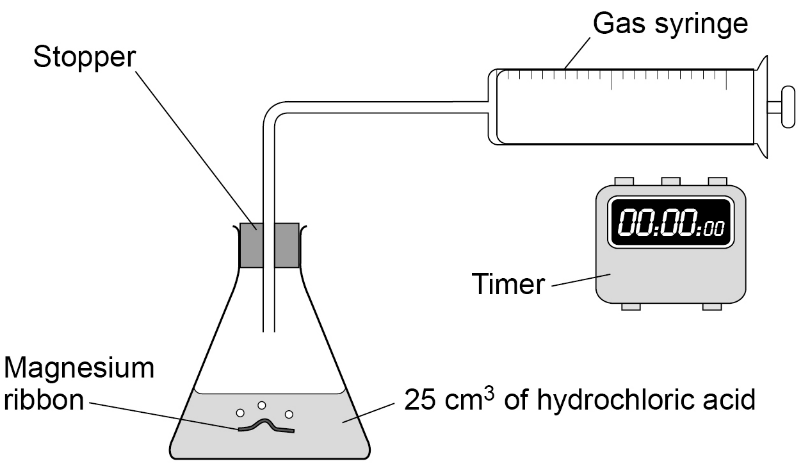
0 4 A student investigated the reaction between magnesium and hydrochloric acid.

The equation for this reaction is:

$$Mg(s) + 2\,HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

Figure 5 shows the apparatus.

Figure 5



This is the method used.

1 Add 25 cm^3 of hydrochloric acid at a temperature of 20 °C to a conical flask.

2 Add 1 cm of magnesium to the conical flask.

3 Insert the stopper and start a timer.

4 Measure the volume of gas collected every 20 seconds for 200 seconds.

5 Repeat steps **1** to **4** using hydrochloric acid at a temperature of 40 °C.

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0 4 . 1 Give **two** observations made when magnesium reacts with hydrochloric acid.

[2 marks]

1 ____

2 ____

0 4 . 2 Give **two control** variables in this investigation.

[2 marks]

1 ____

2 ____

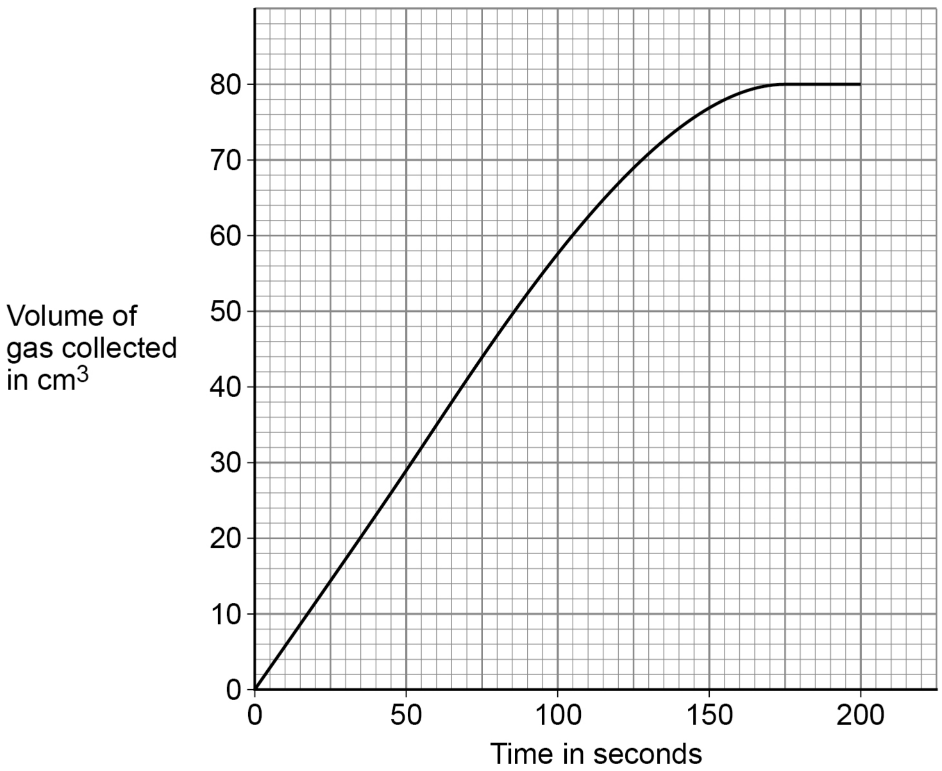
Question 4 continues on the next page

Do not write outside the box

Excess magnesium was reacted with hydrochloric acid at 20 °C.

Figure 6 shows the results.

Figure 6



0 4 . 3 What time did the reaction stop?

Use **Figure 6**.

[1 mark]

_______________ s

0 4 . 4 Why does the reaction stop?

[1 mark]

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____

Do not write outside the box

0 4 . 5 Determine the mean rate of reaction between 0 and 50 seconds.

Give the unit.

Use **Figure 6**.

[4 marks]

Mean rate of reaction = ______________ Unit ______________

0 4 . 6 The student then reacted excess magnesium with hydrochloric acid at 40 °C.

Draw a line on **Figure 6** to show the expected results.

[2 marks]

0 4 . 7 Explain how increasing the temperature of the hydrochloric acid affects the rate of reaction.

[3 marks]

15

Turn over for the next question

Do not write outside the box

0 5 This question is about energy transfers in chemical reactions.

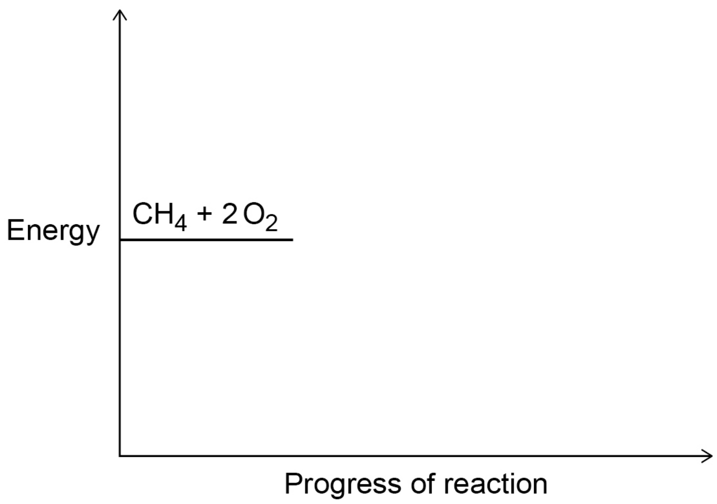
0 5 . 1 The reaction between methane and oxygen is **exothermic**.

The equation for this reaction is:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$$

Figure 7 shows part of an energy level diagram for the reaction between methane and oxygen.

Figure 7



Complete **Figure 7**.

You should label:

- the activation energy
- the overall energy change
- the product line, including the formulae of the products.

[4 marks]

Do not write outside the box

0 5 . 2 The reaction of hydrogen with oxygen is also exothermic.

Figure 8 shows the displayed structures for the reaction between hydrogen and oxygen.

Figure 8

$$2H-H + O=O \longrightarrow 2H-O-H$$

Table 4 shows the bond energies.

Table 4

Bond	Bond energy in kJ/mol
H — H	436
O ═ O	498
O — H	464

Calculate the enthalpy change (ΔH) of the reaction.

Use **Figure 8** and **Table 4**.

[4 marks]

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Enthalpy change (ΔH) = _______________ kJ/mol

Question 5 continues on the next page

0 5 . 3 Methane and hydrogen can be used as fuels.

Table 5 shows some information about methane and hydrogen.

Table 5

Fuel	**Raw material**	**Products of combustion**
Methane	Crude oil	Carbon dioxide and water
Hydrogen	Water	Water

Give **two** advantages of using hydrogen instead of methane as a fuel.

[2 marks]

1 ____

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2 ____

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10

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Turn over for the next question

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0 6 This question is about alcohols.

0 6 . 1 Which is a use of ethanol?

[1 mark]

Tick (✓) **one** box.

As an alloy ☐

As a food colouring ☐

As a solvent ☐

0 6 . 2 A student tested an unknown compound **X**.

The student made the following observations:

- **X** was flammable
- **X** had no reaction with sodium
- when **X** was mixed with water and universal indicator the solution turned green.

The student concluded that compound **X** is ethanol.

Evaluate the student's conclusion.

[3 marks]

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0 6 . 3 Ethanol can be converted into ethanoic acid.

What is the name of this type of reaction?

[1 mark]

Tick (✓) **one** box.

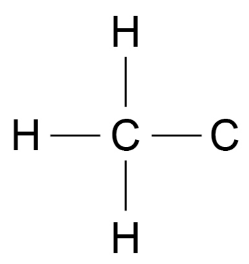
Combustion ☐

Neutralisation ☐

Oxidation ☐

0 6 . 4 Complete the displayed formula of ethanoic acid.

[2 marks]



0 6 . 5 Ethanol reacts with ethanoic acid to produce an ester.

Name the ester produced.

[1 mark]

Question 6 continues on the next page

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0 6 . 6 Industrially, ethanol is produced by reacting ethene with steam.

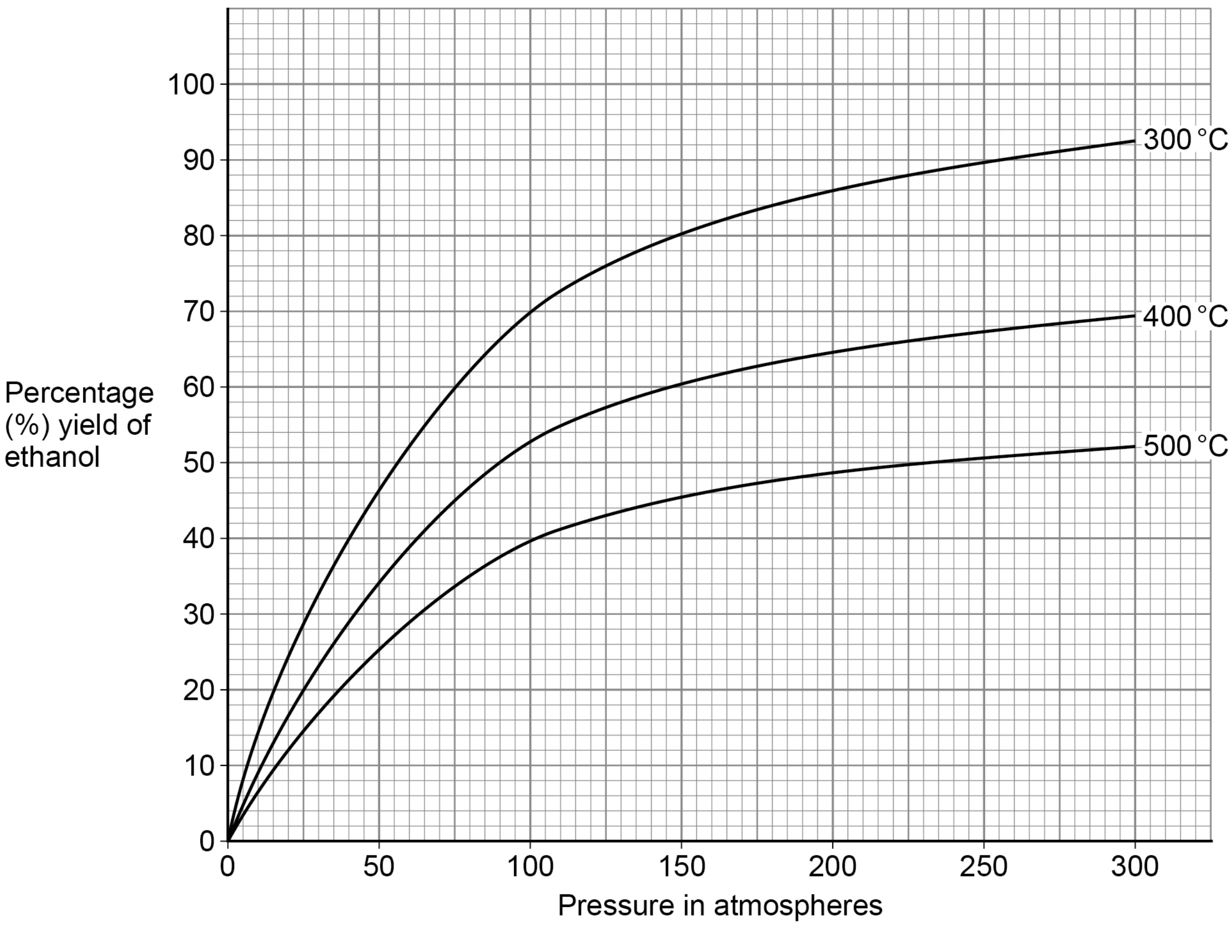
The equation for the reaction is:

$$C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(g)$$

The forward reaction is **exothermic**.

Figure 9 shows the percentage yield of ethanol under different conditions.

Figure 9



In the industrial process the conditions used are:

- a temperature of 300 °C
- a pressure of 65 atmospheres
- a phosphoric acid catalyst.

Explain why these conditions are chosen for the economical production of ethanol from ethene and steam.

You should refer to the rate of reaction and position of equilibrium.

[6 marks]

14

Turn over for the next question

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0 7 This question is about acids and alkalis.

0 7 . 1 Which ion makes solutions acidic?

[1 mark]

Tick (✓) **one** box.

H^+ ☐

Na^+ ☐

OH^- ☐

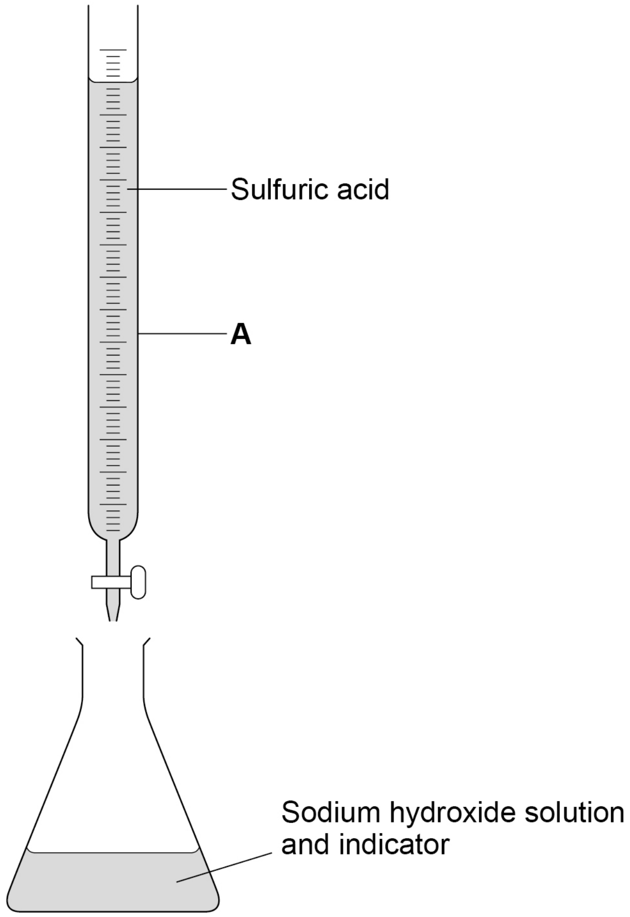
SO_4^{2-} ☐

Question 7 continues on the next page

A student titrated sulfuric acid with sodium hydroxide solution.

Figure 10 shows the apparatus.

Figure 10



This is the method used:

1 Measure 25.0 cm^3 sodium hydroxide solution into a conical flask.

2 Add three drops of indicator solution to the sodium hydroxide solution.

3 Add 1 cm^3 sulfuric acid to the sodium hydroxide solution.

4 Continue adding 1 cm^3 sulfuric acid until the indicator changes colour.

5 Record the total volume of sulfuric acid added.

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0 7 . 2 What is the name of equipment **A** in **Figure 10**?

[1 mark]

Tick (✓) **one** box.

Beaker ☐

Burette ☐

Measuring cylinder ☐

Pipette ☐

0 7 . 3 Give **two** improvements to the method to obtain more accurate results.

[2 marks]

1 ______________________________

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2 ______________________________

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Question 7 continues on the next page

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0 7 . 4 An indicator was added to the sodium hydroxide solution.

Name a suitable indicator to use.

Give the expected colour change at neutralisation.

[2 marks]

Indicator ____

Colour change from ________________________ to ________________________

0 7 . 5 The equation for the reaction is:

$$H_2SO_4(aq) + 2\,NaOH(aq) \rightarrow Na_2SO_4(aq) + 2\,H_2O(l)$$

13.5 cm^3 of 0.20 mol/dm^3 sulfuric acid neutralised 25.0 cm^3 of sodium hydroxide solution.

Calculate the concentration of the sodium hydroxide solution.

[4 marks]

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Concentration of sodium hydroxide solution = ______________ mol/dm^3

10

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Turn over for the next question

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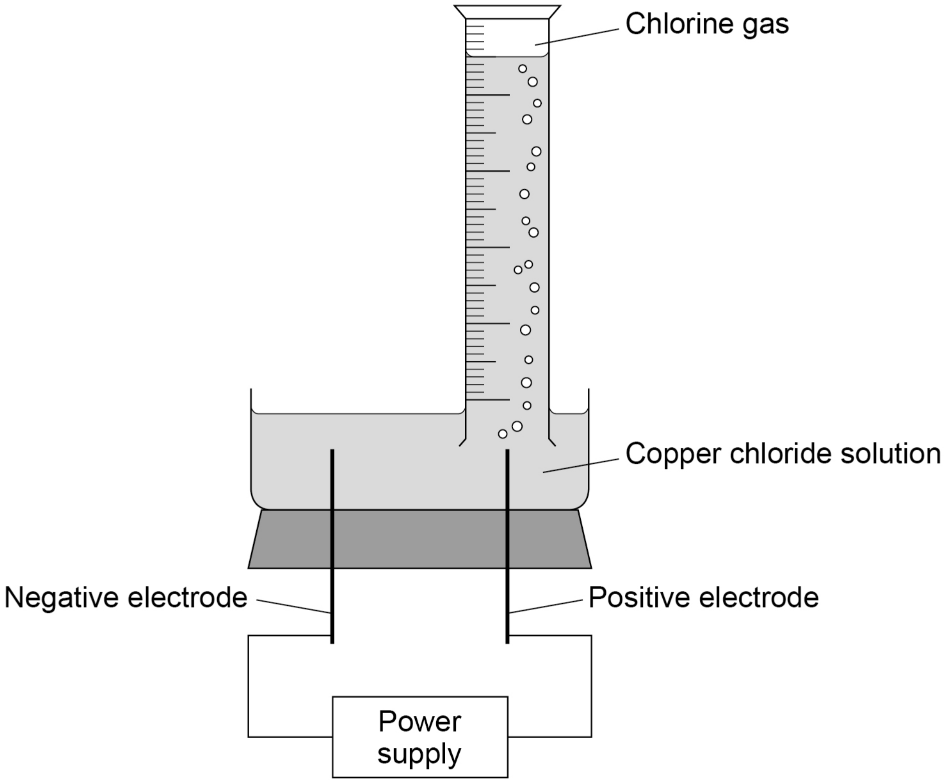
0 8 This question is about electrolysis.

A teacher demonstrated the electrolysis of copper chloride solution.

Figure 11 shows the apparatus.

Figure 11



0 8 . 1 What is the electrolyte in **Figure 11**?

[1 mark]

Tick (✓) **one** box.

Chlorine gas ☐

Copper chloride solution ☐

Positive electrode ☐

Power supply ☐

Do not write outside the box

0 8 . 2 Why does copper chloride need to be dissolved in water for electrolysis to occur?

[1 mark]

0 8 . 3 Explain how copper metal forms at the negative electrode.

[4 marks]

Question 8 continues on the next page

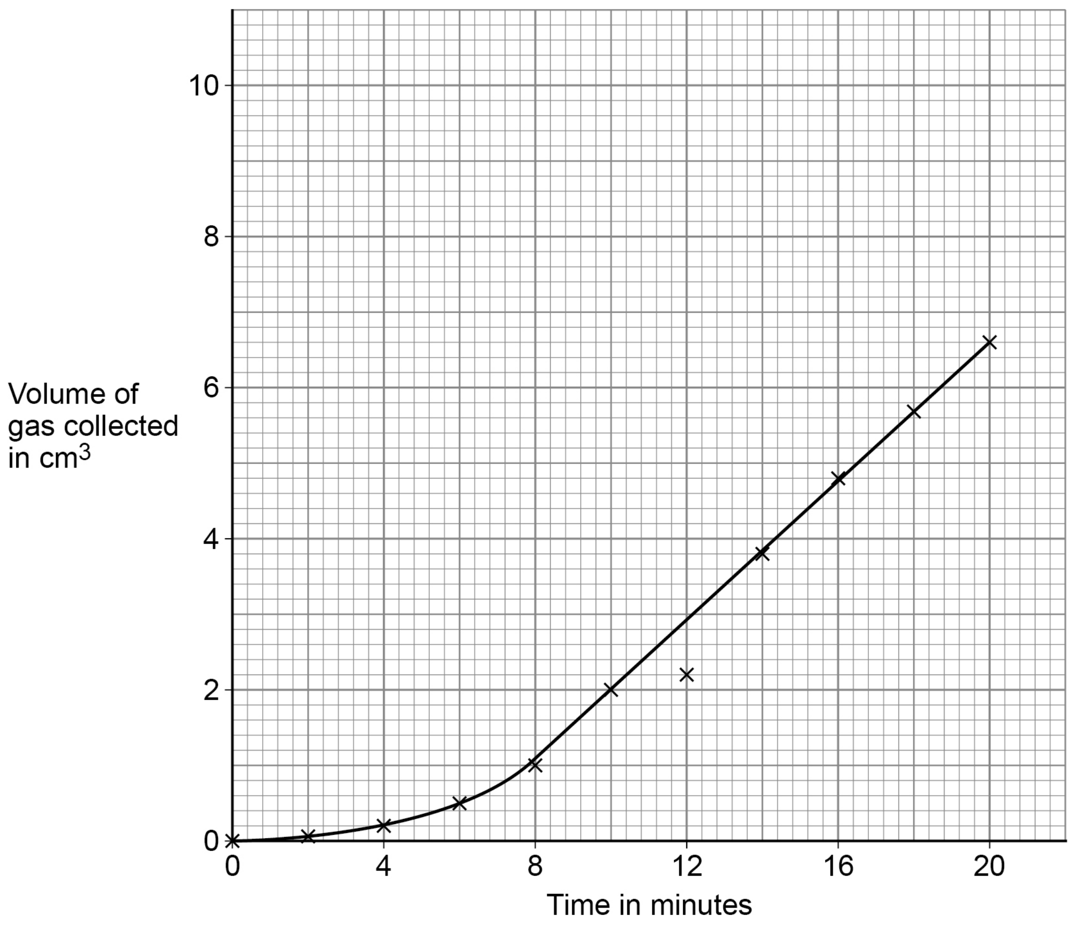
Do not write outside the box

The teacher measured the volume of chlorine gas collected at the positive electrode every 2 minutes.

Chlorine is soluble in water.

Figure 12 shows the results.

Figure 12



0 8 . 4 One of the results shown in **Figure 12** is anomalous.

Which result is anomalous?

Suggest **one** reason for the anomalous result.

[2 marks]

Anomalous result ______________________________

Reason ______________________________

____

Do not write outside the box

0 8 . 5 Explain the trends shown in **Figure 12**.

[4 marks]

0 8 . 6 Determine the number of moles of gas collected after 16 minutes.

The volume of 1 mole of any gas at room temperature and pressure is 24 dm^3.

Use **Figure 12**.

[4 marks]

Number of moles of gas collected = ____________

16

END OF QUESTIONS

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